

RTL2GDSII Workshop Project

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Project title:

- a) Design an 8-bit shifter which can shift left or right by N-bits, based on the given inputs using the Verilog HDL
- b) Generate GDS-II layout of the design using Synopsys flow scripts.
 - i. Report Design Area
 - ii. Clock frequency, where slack value is positive but less than 1.
 - iii. Report power consumption

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Q)

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b) Generate GDS-II layout of the design using Synopsys flow scripts.

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ii. Clock frequency, where slack value is positive but less than 1.

iii. Report power consumption

a) RTL Code of design and testbench in Verilog RTL

To open and create file of design using linux terminal command in rtl folder:

vi shifter_rtl.v

do the same to create .v file of testbench using linux terminal command in rtl file:

vi shifter_tb.v

type i for writing in text file

type :w for saving the text file

type :q to exit from text file

1. Design Code : (shifter_rtl.v file)

```
module shifter_8bit (  
    input [7:0] data_in,  
    input [2:0] shift_amt,  
    input dir,          // 0 = left, 1 = right  
    input Clock,  
    output reg [7:0] data_out  
);  
  
// Internal registers  
reg [7:0] reg_data;  
reg [2:0] reg_shift;  
reg reg_dir;  
reg [7:0] shifted_val;  
  
// Input latching  
always @(posedge Clock) begin  
    reg_data <= data_in;
```

```

    reg_shift <= shift_amt;
    reg_dir <= dir;
end

// Output latching
always @(posedge Clock) begin
    data_out <= shifted_val;
end

// Shift computation
always @(*) begin
    if (reg_dir == 1'b0)
        shifted_val = reg_data << reg_shift; // Left shift
    else
        shifted_val = reg_data >> reg_shift; // Right shift
end
endmodule

```

2. Testbench Code: (shifter_tb.v file)

```

`timescale 1ns / 1ps
`include "shifter_rtl.v" // includes the module definition

module tb_shifter_8bit;

    reg [7:0] data_in;
    reg [2:0] shift_amt;
    reg dir;
    reg Clock;
    wire [7:0] data_out;

```

```

// Instantiate the design
shifter_8bit uut (
    .data_in(data_in),
    .shift_amt(shift_amt),
    .dir(dir),
    .Clock(Clock),
    .data_out(data_out)
);

// Clock generation
initial begin
    Clock = 0;
    forever #5 Clock = ~Clock; // 10 ns clock period
end

// Test sequence
initial begin
    // VCD waveform generation
    $fsdbDumpvars();
    $display("Time | DIR | SHIFT_AMT | DATA_IN | DATA_OUT");
    $monitor("%4t | %b | %d | %b | %b", $time, dir, shift_amt, data_in,
data_out);

    // Test case 1
    data_in = 8'b10110011;
    shift_amt = 3;
    dir = 0; // Left
    #10;

```

```
// Test case 2  
data_in = 8'b10110011;  
shift_amt = 3;  
dir = 1; // Right  
#10;
```

```
// Test case 3  
data_in = 8'b11110000;  
shift_amt = 1;  
dir = 0;  
#10;
```

```
// Test case 4  
data_in = 8'b00001111;  
shift_amt = 2;  
dir = 1;  
#10;
```

```
// Test case 5  
data_in = 8'b11001100;  
shift_amt = 7;  
dir = 0;  
#10;
```

```
$finish;
```

```
end
```

```
endmodule
```

Compile RTL design file using this linux terminal command:

vcs -full64 shifter_rtl.v -debug_access+all -lca -kdb

Compile Testbench file using this linux terminal command:

vcs -full64 shifter_tb.v -debug_access+all -lca -kdb

To run simulation in Verdi , command is:

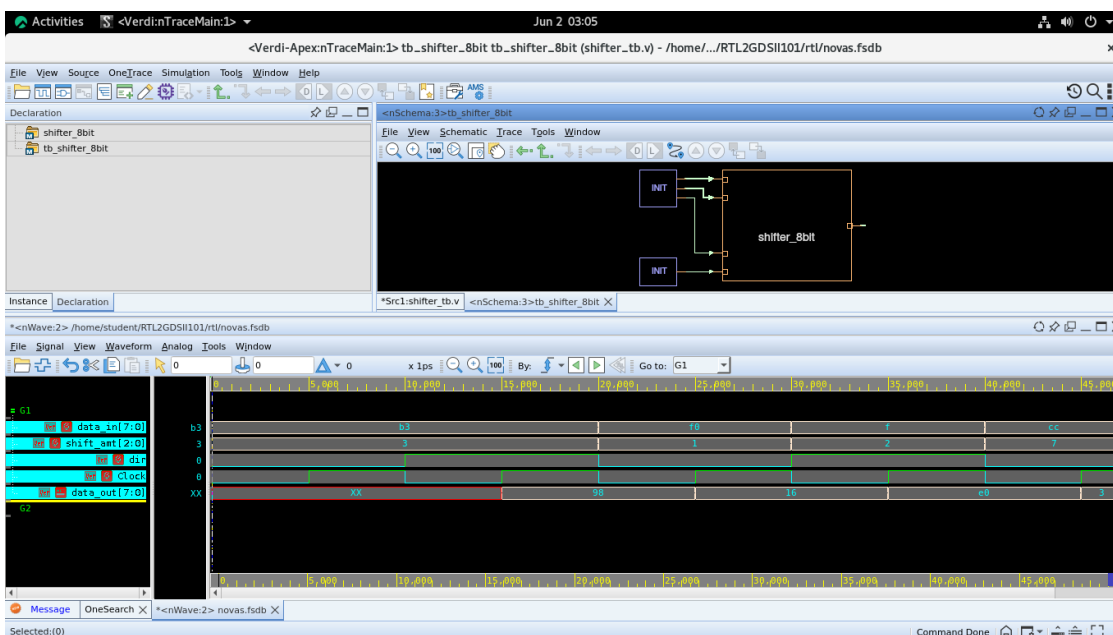
./simv Verdi

```
Activities Terminal Jun 2 03:03
student@ict-chipin:~/RTL2GDSII101/rtl

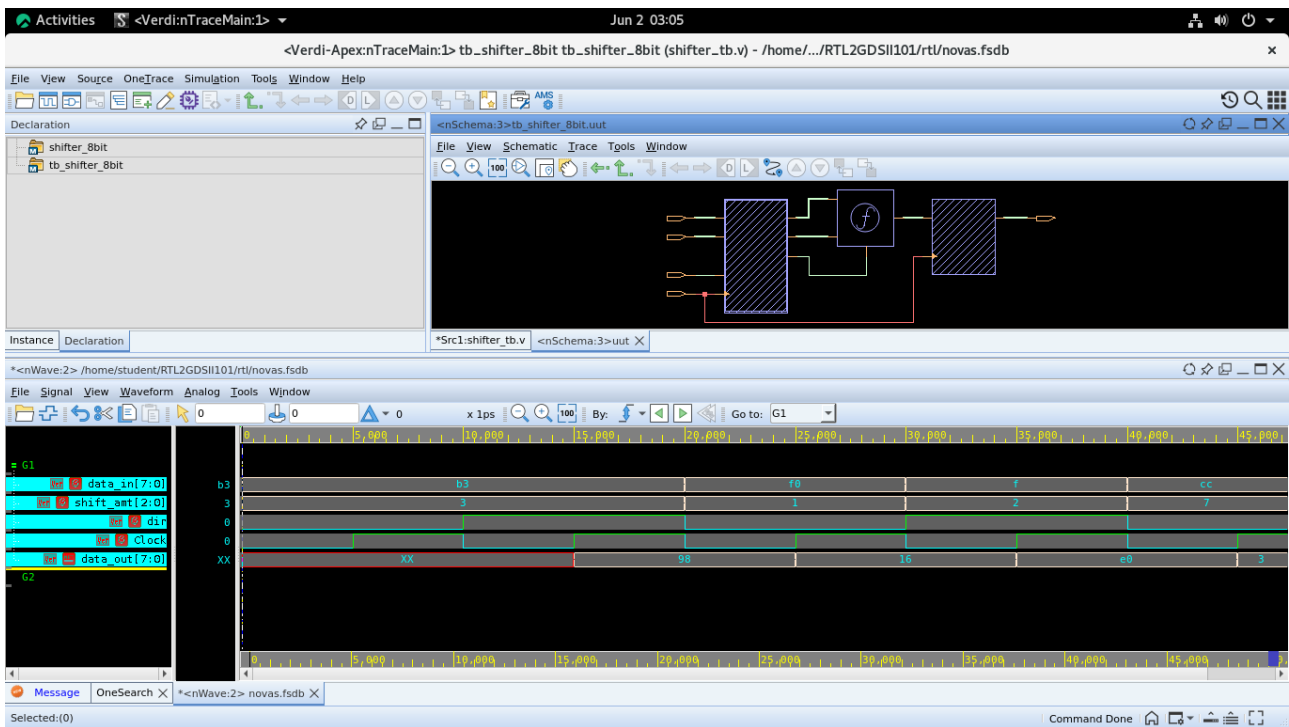
File Edit View Search Terminal Help
Please use '-no save' simv switch to avoid re-execution or '-suppress=ASLR_DETECTED_INFO' to suppress this message.
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09_Full64; Runtime version W-2024.09_Full64; Jun 2 03:02 2025
[SCL] SCL trace is enabled.
WARNING:Hostname to IPv4 and IPv6 address lookup failed for the host synopsys
Checkout succeeded: VCS-BASE-RUNTIME/AF3AD4D7787CF4828637
License file: 27020@c2s.cdacb.in
License Server: 27020@c2s.cdacb.in
[SCL] 06/02/2025 03:02:31 Checking status for feature VCS-BASE-RUNTIME
[SCL] 06/02/2025 03:02:31 PID:18886 Client:ict-chipin.sot.pdpu.ac.in Server:27020@c2s.cdacb.in Checkout succeeded VCS-BASE-RUNTIME 2024.09 from VCS-Ba
se-RunTime-Pkg
*Verdi* Loading libsscore vcs202409.so
FSDB Dumper for VCS, Release Verdi_W-2024.09, Linux x86_64/64bit, 08/17/2024
(C) 1996 - 2024 by Synopsys, Inc.
*Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the programs that are using this file.
*Verdi* : Create FSDB file 'novas.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : End of traversing.
Time | DIR | SHIFT_AMT | DATA_IN | DATA_OUT
0 | 0 | 3 | 10110011 | xxxxxxxx
10000 | 1 | 3 | 10110011 | xxxxxxxx
15000 | 1 | 3 | 10110011 | 10011000
20000 | 0 | 1 | 11110000 | 10011000
25000 | 0 | 1 | 11110000 | 00010110
30000 | 1 | 2 | 00001111 | 00010110
35000 | 1 | 2 | 00001111 | 11100000
40000 | 0 | 7 | 11001100 | 11100000
45000 | 0 | 7 | 11001100 | 00000011
$finish called from file "shifter_tb.v", line 64.
$finish at simulation time 50000
VCS Simulation Report
Time: 50000 ps
CPU Time: 0.200 seconds; Data structure size: 0.0Mb
Mon Jun 2 03:02:34 2025
[SCL] 06/02/2025 03:02:34 PID:18886 Client:ict-chipin.sot.pdpu.ac.in checkin (null)
[student@ict-chipin rtl]$
```

To view waveform in Verdi :

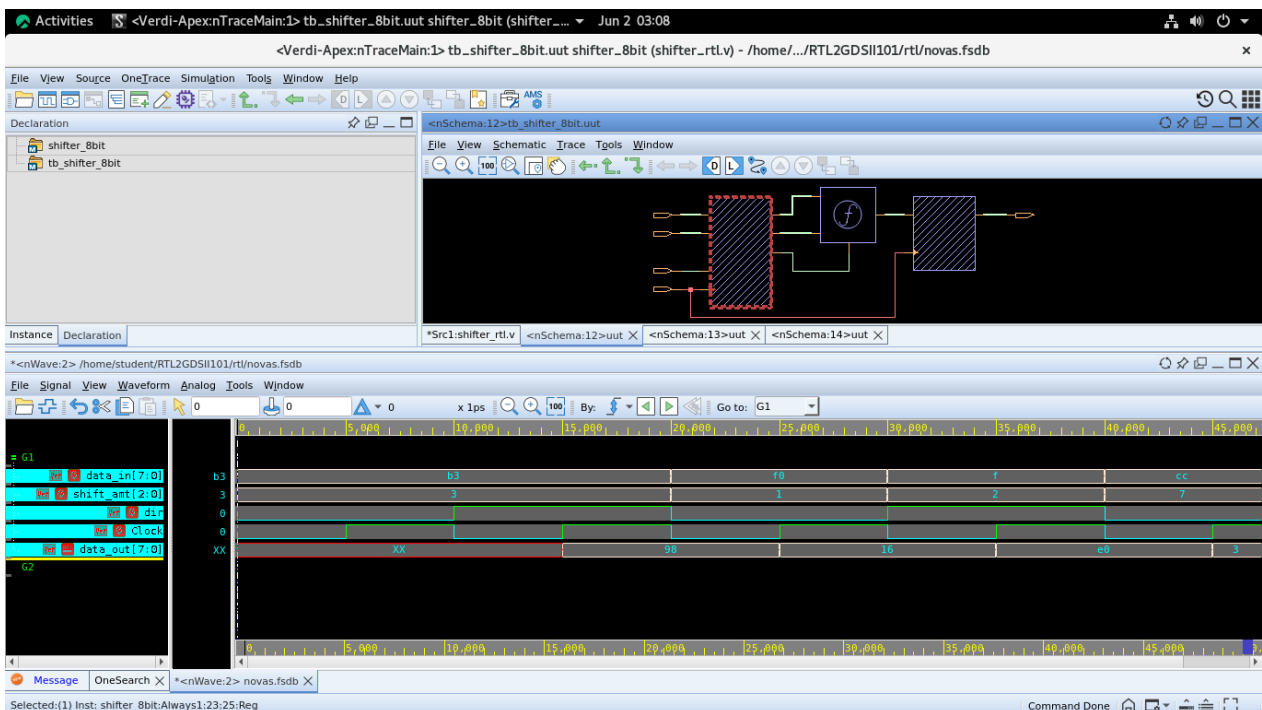
verdi -ssf novas.fsdb -nologo



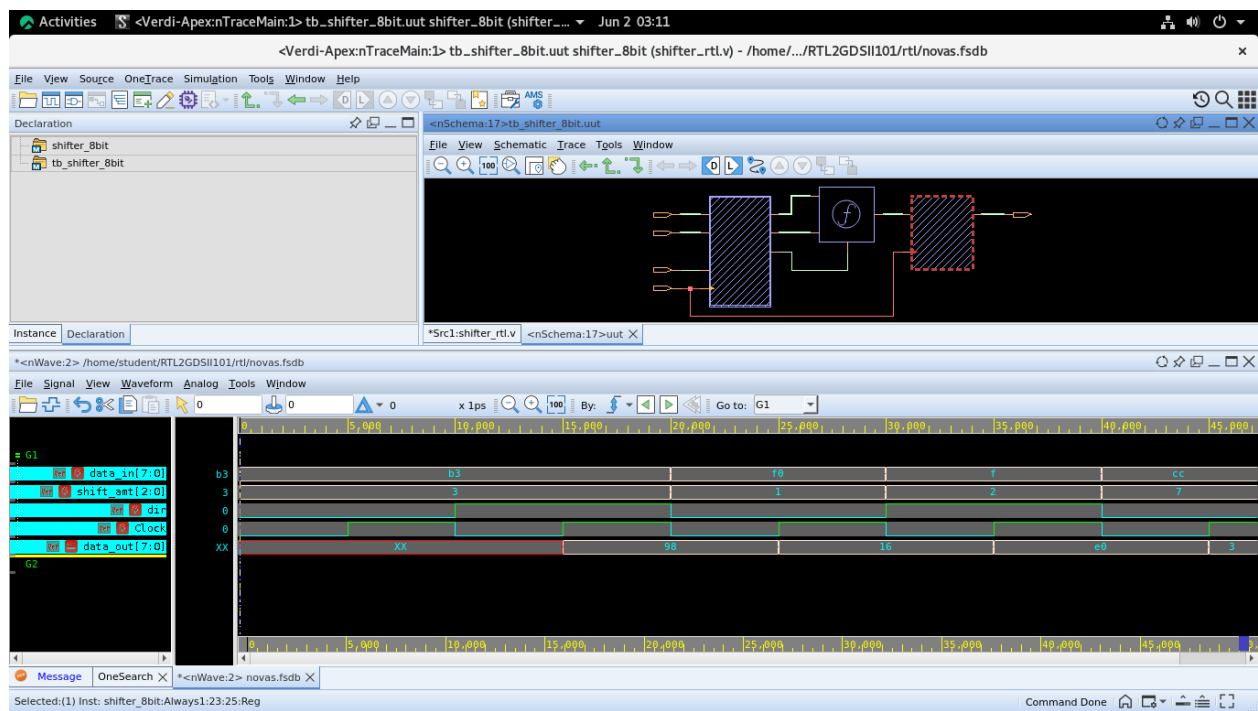
Inside shifter_8bit block :



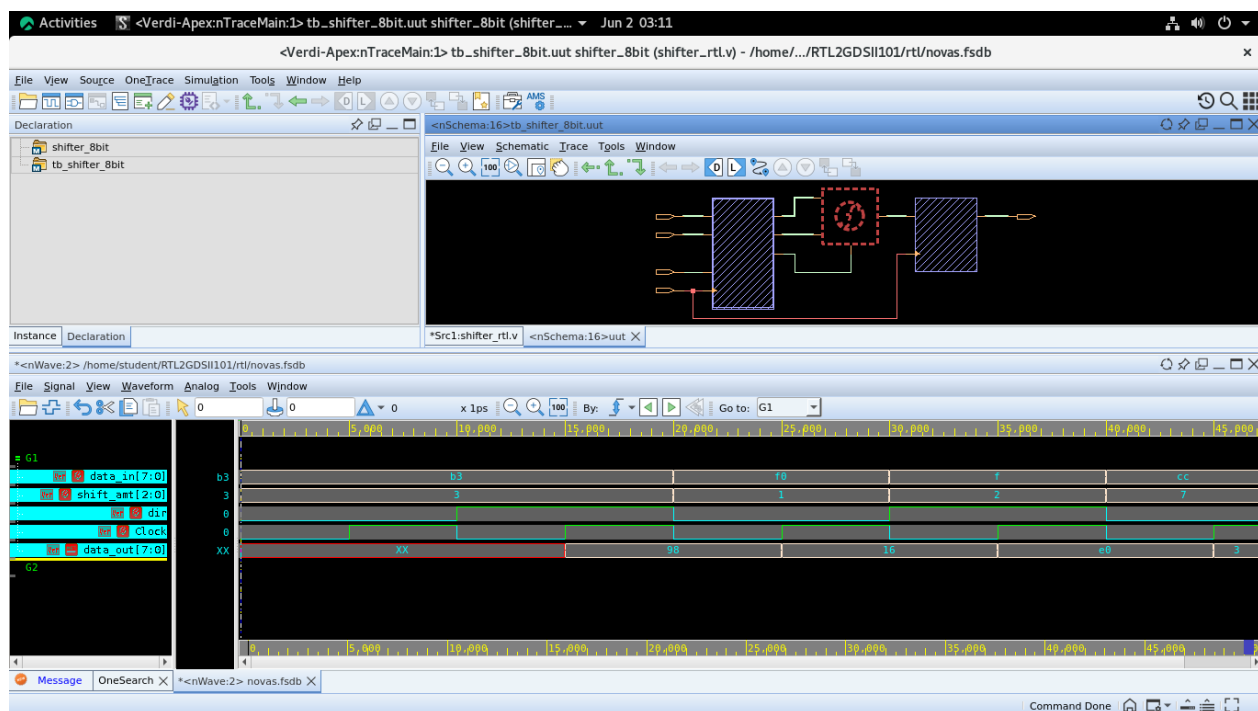
the red boundary box below is showing input latching block :



The red boundary box below is showing output latching block:



The red boundary box below is showing shift computation block:



Logic Synthesis:

For Logic synthesis i used following standard library located in ref/lib/stdcell_rvt:

saed32rvt_tt0p78vn40c.db

do this change in DC/rm_setup/common_setup.tcl and change on line 34 of this file :

```
set TARGET_LIBRARY_FILES "$PDK_PATH/lib/stdcell_rvt/saed32rvt_tt0p78vn40c.db" ;
```

In DC, run_dc.tcl is a main TCL script for synthesis

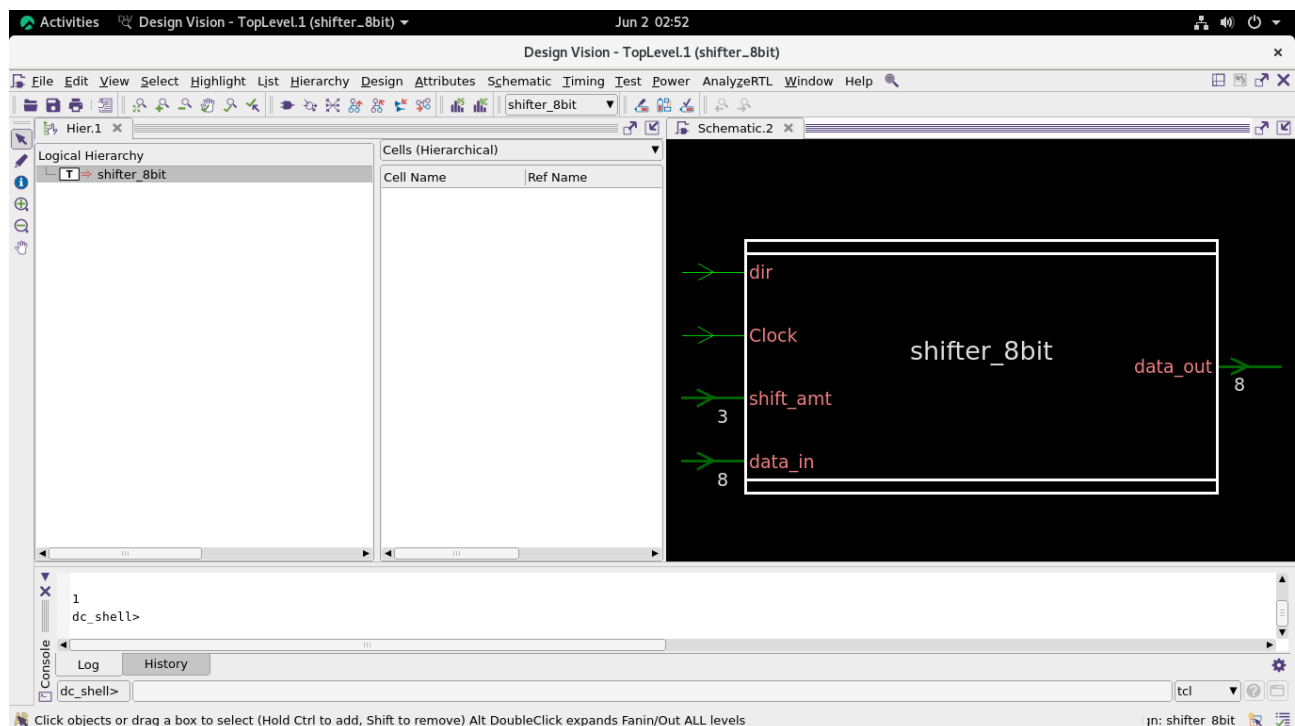
uncomment or modify lines to choose which circuit to synthesize (e.g., full adder, mux, etc.) by opening run_dc.tcl using command → vi run_dc.tcl

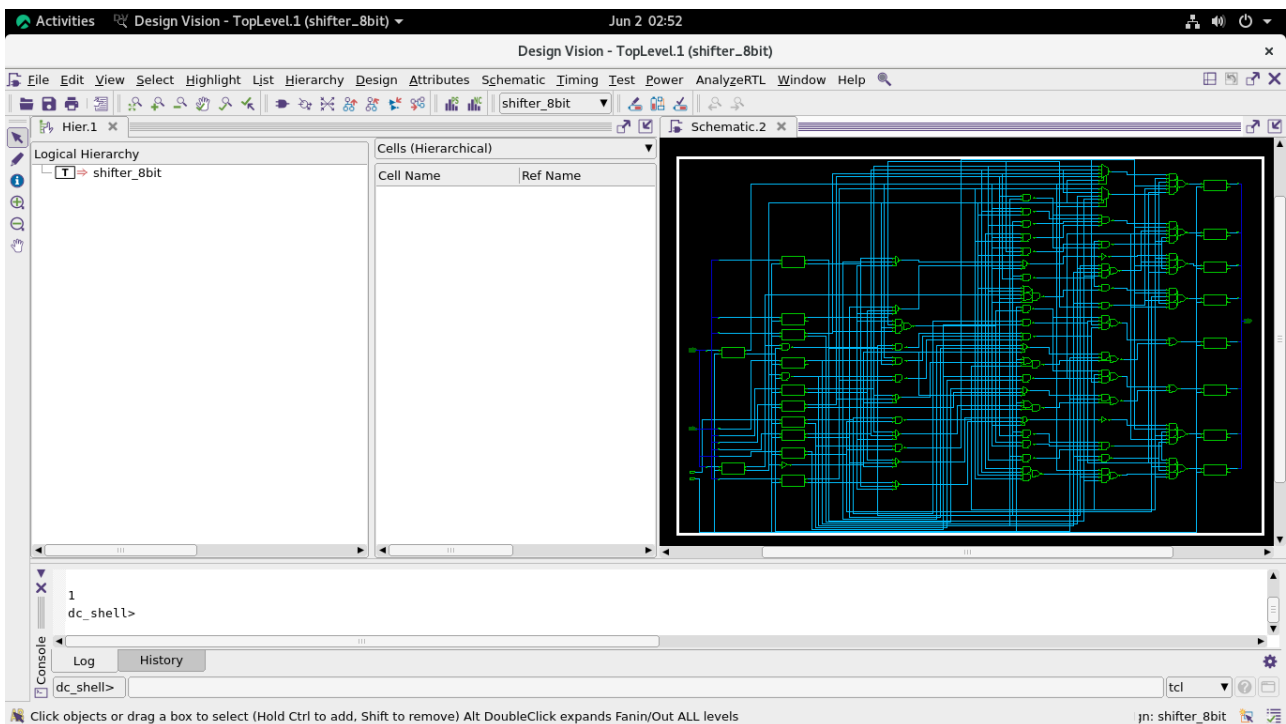
I didn't uncomment any lines and I did ultra compile.

Open in DC directory and type dc_shell to initiate DC compiler.

To run synthesis script → source run_dc.tcl (remember to be in DC file directory for this command)

Now to open DC Compiler GUI → start_gui





We can also modify timing constraints in .sdc file located in CONSTRAINTS:

create_clock -period 1.3 [get_ports Clock]

I put 1.3 (originally it was 1)

Now analyse results:

Type in dc_shell → report_timing

```

student@ict-chipin:~/RTL2GDSII101/DC
shifter_8bit      8000      saed32rvtt0p78vn40c

Point              Incr      Path
-----
clock Clock (rise edge)      0.00      0.00
clock network delay (ideal)  0.00      0.00
reg_shift_reg[0]/CLK (DFFX1_RVT)  0.00      0.00 r
reg_shift_reg[0]/QN (DFFX1_RVT)  0.19      0.19 f
U64/Y (INVX0_RVT)      0.14      0.34 r
U110/Y (MUX21X1_RVT)  0.22      0.55 f
U70/Y (MUX21X1_RVT)  0.14      0.70 f
U62/Y (INVX0_RVT)      0.06      0.75 r
U112/Y (A0222X1_RVT)  0.14      0.90 r
data_out_reg[4]/D (DFFX1_RVT)  0.01      0.91 r
data arrival time      0.91

clock Clock (rise edge)      1.30      1.30
clock network delay (ideal)  0.00      1.30
clock uncertainty      -0.30      1.00
data_out_reg[4]/CLK (DFFX1_RVT)  0.00      1.00 r
library setup time      -0.09      0.91
data required time      0.91

data required time      0.91
data arrival time      -0.91
-----
slack (MET)              0.00

Writing verilog file '/home/student/RTL2GDSII101/DC/results/shifter_8bit.mapped.v'.
1
Current design is 'shifter_8bit'.
dc_shell> report_timing
*****

```

We need slack to be 0, it could be greater than 0 or could be less than 0.

At 0 and greater than 0 we get (MET) and less than 0 (for -ve values) we get (VIOLATED)

Slack 0 means that design is perfectly timed and its uses all the available time (eg. Clock period) without violating constraints which is efficient and safe.

To export the synthesized design we need to uncomment one line in run_dc.tcl file:

```
write -format verilog -hierarchy -output ../results/shifter_8bit.mapped.v
```

shifter_8bit.mapped.v → this is synthesized netlist file which contains gate-level synthesized Verilog of my design. We can know gates used for our design using this in detail without looking at circuit diagram. This file is in DC/results.

Generating GDS2 layout in IC Compiler 2 (ICCII) using flowscripts :

First we open ICC2 directory, in this directory we can see scripts, in scripts there are 5 tcl files which should be executed in following order :

1. floorplan.tcl
2. power_planning.tcl
3. placement.tcl
4. clock.tcl
5. route.tcl

When we open floorplan.tcl, we can first 3 lines:

set PDK_PATH ../../ref → define PDK (Process Design kit) location to ref folder so that other commands can refer to it using the variable \$PDK_PATH

create_lib -ref_lib \$PDK_PATH/lib/ndm/saed32rvt_c.ndm SHIFTER_LIB →

To load the standard cell library (which includes logic gates, flip-flops, etc.) needed for understanding and placing the design.

read_verilog {../../DC/results/shifter_8bit.mapped.v} -library SHIFTER_LIB -design shifter_8bit -top shifter_8bit → to import your mapped design (after synthesis) for further steps like floor planning, placement, and routing.

Open ICCII shell using command: **icc2_shell** (remember-do this in ICCII folder)

We can run these three commands line by line in icc2_shell terminal or we can simply source the file in icc2_shell terminal : **source scripts/floorplan.tcl**

In this file we can see 10 scenarios which gives floorplan layout of the design.

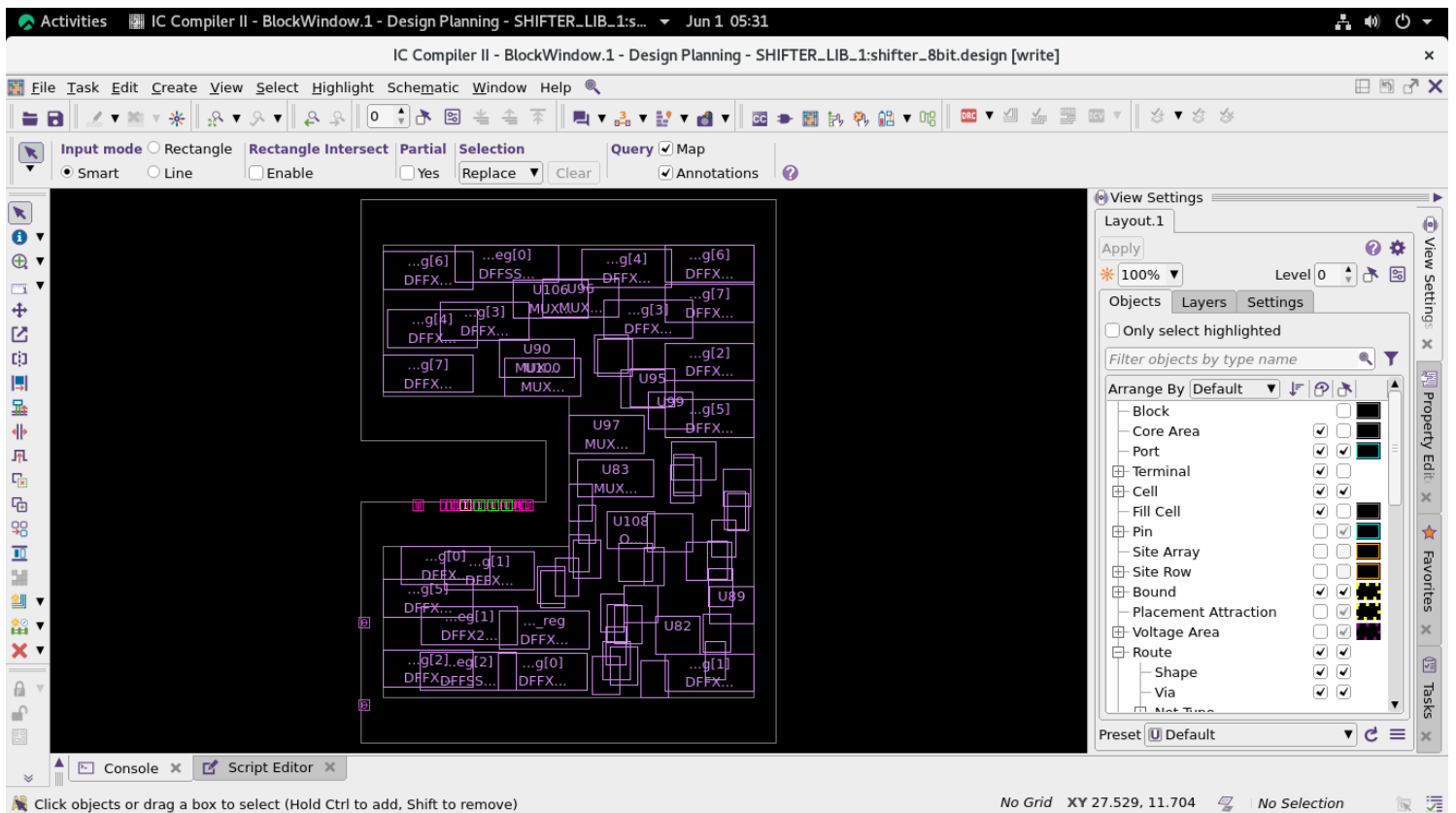
We comment or uncomment whatever scenario we want to try and source the floorplan file again.

We can also just copy the scenario we want and paste in icc2_shell and run.

create_placement –floorplan

create_placement – floorplan

I got this layout diagram:



Following is qor report after floorplan using command: **report_qor**

Cell area = 305.23, combinational cell count = 67

Leaf cell count = 87, Buffers cell count = 7, Inverter cell count = 3

```
student@ict-chipin:~/RTL2GDSII101/IC11
Cell Count
-----
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 87
Buf/Inv Cell Count: 10
Buf Cell Count: 7
Inv Cell Count: 3
Combinational Cell Count: 67
  Single-bit Isolation Cell Count: 0
  Multi-bit Isolation Cell Count: 0
  Isolation Cell Banking Ratio: 0.00%
  Single-bit Level Shifter Cell Count: 0
  Multi-bit Level Shifter Cell Count: 0
  Level Shifter Cell Banking Ratio: 0.00%
  Single-bit ELS Cell Count: 0
  Multi-bit ELS Cell Count: 0
  ELS Cell Banking Ratio: 0.00%
Sequential Cell Count: 20
  Integrated Clock-Gating Cell Count: 0
  Sequential Macro Cell Count: 0
  Single-bit Sequential Cell Count: 20
  Multi-bit Sequential Cell Count: 0
  Sequential Cell Banking Ratio: 0.00%
  BitsPerFlop: 1.00
Macro Count: 8
Area
-----
Combinational Area: 172.56
Noncombinational Area: 132.66
Buf/Inv Area: 23.89
Total Buffer Area: 20.08
Total Inverter Area: 3.81
Macro/Block Box Area: 0.00
Net Area: 0
Net XLength: 1025.76
Net YLength: 1413.60
Cell Area (netlist): 305.23
Cell Area (netlist and physical only): 305.23
Net Length: 2439.36
Design Rules
-----
Total Number of Nets: 112
Nets with Violations: 13
Max Trans Violations: 13
Max Cap Violations: 0
```

Following is timing report after floorplan using command : **report_timing**

Slack = 0.37

```
student@ict-chipin:~/RTL2GDSII101/IC11
[icc2-lic Mon Jun 2 01:31:39 2025] Sending check-out request for 'ICCompilerII-NX' (1) with wait option
[icc2-lic Mon Jun 2 01:31:39 2025] Checking status for feature ICCompilerII-NX
[icc2-lic Mon Jun 2 01:31:39 2025] PID:8383 Client:ict-chipin.sot.pdpu.ac.in Server:27020@c2s.cdacb.in Checkout succeeded ICCompilerII-NX 2024.09
[icc2-lic Mon Jun 2 01:31:39 2025] Check-out request for 'ICCompilerII-NX' with wait option succeeded
[icc2-lic Mon Jun 2 01:31:39 2025] Sending checkout check request for 'ICCompilerII-NX'
[icc2-lic Mon Jun 2 01:31:39 2025] Checkout check request for 'ICCompilerII-NX' returned 0
[icc2-lic Mon Jun 2 01:31:39 2025] Sending count request for 'ICCompilerII-NX'
[icc2-lic Mon Jun 2 01:31:39 2025] Count request for 'ICCompilerII-NX' returned 1
[icc2-lic Mon Jun 2 01:31:39 2025] Check-out of alternate set of keys directly with queueing was successful
Report : timing
-path type full
-delay type max
-max_paths 1
-report by design
Design : shifter_8bit
Version: W-2024.09
Date : Mon Jun 2 01:31:39 2025
Startpoint: data_out_reg[5] (rising edge-triggered flip-flop clocked by Clock)
Endpoint: data_out[5] (output port clocked by Clock)
Mode: func
Corner: nom
Scenario: func:nom
Path Group: **reg2out_default**
Path Type: max
Point          Incr      Path
-----
clock Clock (rise edge)          0.00      0.00
clock network delay (propagated) 0.07      0.07
data_out_reg[5]/CLK (DFFX1_RVT) 0.00      0.07 r
data_out_reg[5]/Q (DFFX1_RVT) 0.09      0.16 f
data_out[5] (out) 0.00      0.16 f
data arrival time 0.16
clock Clock (rise edge)          1.30      1.30
clock network delay (propagated) 0.04      1.34
clock uncertainty -0.30      1.04
output external delay -0.50      0.54
data required time 0.54
data arrival time -0.16
slack (MET) 0.37
```

Following is the power consumption report after floorplan using command :

report_power

```
Activities Terminal Jun 2 01:32
student@ict-chipin:~/RTL2GDSI101/ICCI

File Edit View Search Terminal Tabs Help

student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x

Temperature: 125.00
Voltage Unit : 1V
Capacitance Unit : 1fF
Time Unit : ns
Temperature Unit : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[7] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell reg_data_reg[7] for parameter Cout. Lowest table value = 0.000100, highest table value = 0.000000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[6] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[5] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[4] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[3] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[2] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[1] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell reg_data_reg[0] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.499992 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell reg_data_reg[0] for parameter Cout. Lowest table value = 0.000100, highest table value = 0.000000, value = 0.000000 (POW-046)
Note - message 'POW-046' limit (10) exceeded. Remainder will be suppressed.

Cell Internal Power = 1.80e+08 pW ( 89.2%)
Net Switching Power = 2.18e+07 pW ( 10.8%)
Total Dynamic Power = 2.02e+08 pW (100.0%)

Cell Leakage Power = 7.52e+08 pW

Attributes
-----
u - User defined power group
l - Includes clock pin internal power

Power Group Internal Power Switching Power Leakage Power Total Power ( % ) Attrs
-----
io_pad 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
memory 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
black_box 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
clock_network 2.04e+08 1.31e+07 9.75e+07 3.15e+08 ( 33.0%) l
register -3.40e+07 3.10e+06 4.13e+08 3.82e+08 ( 40.1%)
sequential 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
combinational 1.02e+07 5.57e+06 2.42e+08 2.57e+08 ( 27.0%)
-----
Total 1.80e+08 pW 2.18e+07 pW 7.52e+08 pW 9.54e+08 pW
l
icc2_shell>
```

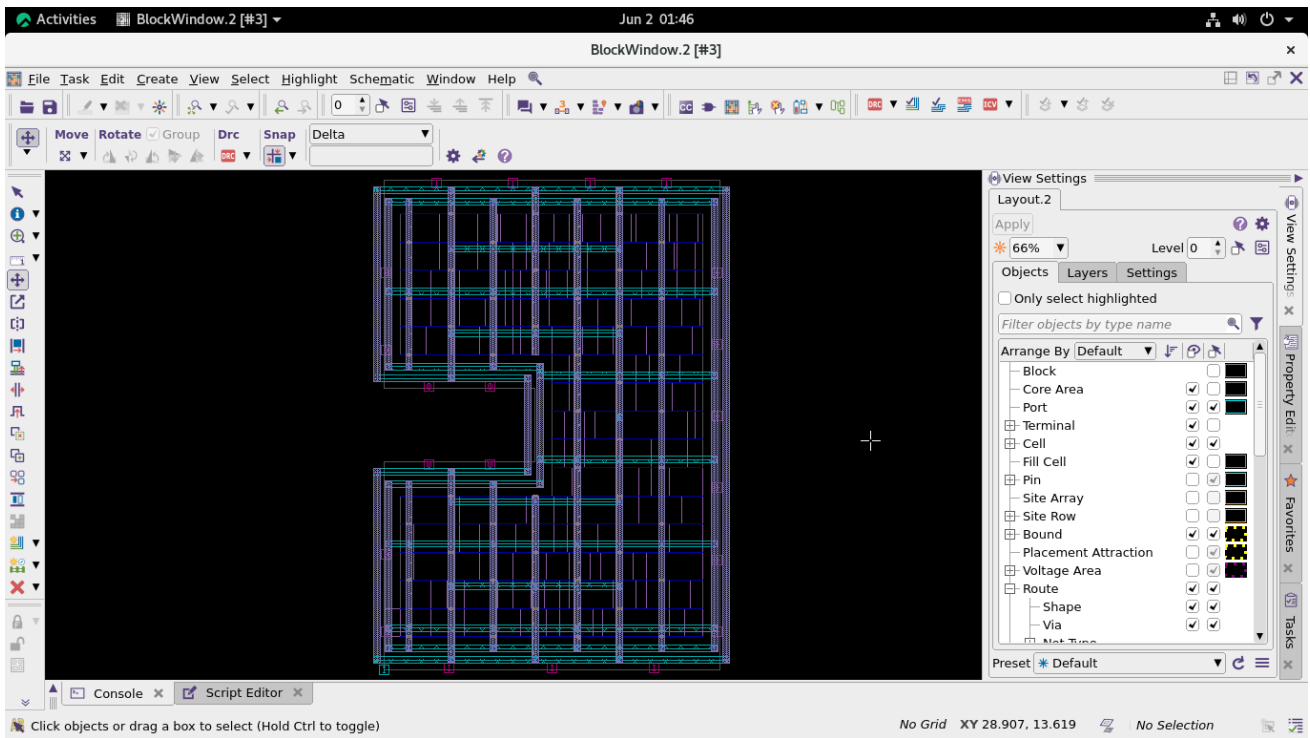
In the layout above we can see the blocks are not arranged properly. This will be fixed after using placement.tcl file.

Now first we need to do power planning which is done by running power_planning.tcl

File in icc2_shell terminal: **source scripts/power_planning.tcl**

Then we will also run placement.tcl file so the blocks will be arranged properly.

So in icc2_shell terminal: **source scripts/placement.tcl**



Here we can see that blocks are arranged properly and sky blue color power rails and dark blue color rails.

Following is qor report after power planning using command: **report_qor**

Cell area = 305.23, combinational cell count = 67

Leaf cell count = 87, Buffers cell count = 7, Inverter cell count = 3

```

Jun 2 01:39
student@ict-chipin:~/RTL2GDSII101/IC11
File Edit View Search Terminal Tabs Help
student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi...
Cell Count
-----
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 87
Buf/Inv Cell Count: 10
Buf Cell Count: 7
Inv Cell Count: 3
Combinational Cell Count: 67
Single-bit Isolation Cell Count: 0
Multi-bit Isolation Cell Count: 0
Isolation Cell Banking Ratio: 0.00%
Single-bit Level Shifter Cell Count: 0
Multi-bit Level Shifter Cell Count: 0
Level Shifter Cell Banking Ratio: 0.00%
Single-bit ELS Cell Count: 0
Multi-bit ELS Cell Count: 0
ELS Cell Banking Ratio: 0.00%
Sequential Cell Count: 20
Integrated Clock-Gating Cell Count: 0
Sequential Macro Cell Count: 0
Single-bit Sequential Cell Count: 0
Multi-bit Sequential Cell Count: 0
Sequential Cell Banking Ratio: 0.00%
BitsPerFlop: 1.00
Macro Count: 0
Area
-----
Combinational Area: 172.56
Noncombinational Area: 132.66
Buf/Inv Area: 23.89
Total Buffer Area: 20.08
Total Inverter Area: 3.81
Macro/Block Box Area: 0.00
Net Area: 0
Net XLength: 1025.76
Net YLength: 1413.68
Cell Area (netlist): 305.23
Cell Area (netlist and physical only): 305.23
Net Length: 2439.36
Design Rules
-----
Total Number of Nets: 112
Nets with Violations: 13
Max Trans Violations: 13
Max Cap Violations: 0

```

Following is timing report after power planning using command : **report_timing**

Slack = 0.37

```
Report : timing
  -path_type full
  -delay_type max
  -max_paths 1
  -report_by design
Design : shifter_8bit
Version: W:2024.09
Date : Mon Jun 2 01:39:46 2025
-----
Startpoint: data_out_reg[5] (rising edge-triggered flip-flop clocked by Clock)
Endpoint: data_out[5] (output port clocked by Clock)
Mode: func
Corner: nom
Scenario: func::nom
Path Group: **reg2out_default**
Path Type: max
-----
Point                               Incr      Path
-----
clock Clock (rise edge)              0.00      0.00
clock network delay (propagated)      0.07      0.07
data_out_reg[5]/CLK (DFFX1_RV1)      0.00      0.07 f
data_out_reg[5]/Q (DFFX1_RV1)        0.00      0.16 f
data_out[5] (out)                    0.00      0.16 f
data arrival time                     0.16
-----
clock Clock (rise edge)              1.30      1.30
clock network delay (propagated)      0.84      1.34
clock uncertainty                     -0.30      1.04
output external delay                 -0.50      0.54
data required time                    0.54
-----
data required time                    0.54
data arrival time                     -0.16
-----
slack (MET)                          0.37
-----
1
lcc2_shell>
```

Following is the power consumption report after power planning using command :

report_power

```
Report : power
  -significant_digits 2
Design : shifter_8bit
Version: W:2024.09
Date : Mon Jun 2 01:48:35 2025
-----
Information: Activity for scenario func::nom was cached, no propagation required. (POW-605)
Mode: func
Corner: nom
Scenario: func::nom
Voltage: 0.85
Temperature: 125.00
-----
Voltage Unit : 1V
Capacitance Unit : 1ff
Time Unit : 1ns
Temperature Unit : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW
-----
Switched supply net power scaling:
scaling for Leakage power
-----
Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)
-----
Cell Internal Power = 1.88e+08 pW ( 89.2%)
Net Switching Power = 2.18e+07 pW ( 10.8%)
Total Dynamic Power = 2.02e+08 pW (100.0%)
Cell Leakage Power = 7.52e+08 pW
-----
Attributes
-----
U - User defined power group
I - Includes clock pin internal power
-----
Power Group   Internal Power   Switching Power   Leakage Power   Total Power   ( % )   Attrs
-----
io_pad        0.00e+00          0.00e+00          0.00e+00        0.00e+00     ( 0.0%)
memory        0.00e+00          0.00e+00          0.00e+00        0.00e+00     ( 0.0%)
black_box     0.00e+00          0.00e+00          0.00e+00        0.00e+00     ( 0.0%)
clock_network 2.04e+08          1.31e+07          9.75e+07        3.15e+08     ( 33.0%)
register      -3.40e+07          3.10e+06          4.13e+08        3.82e+08     ( 40.1%)
sequential    0.00e+00          0.00e+00          0.00e+00        0.00e+00     ( 0.0%)
combinational 1.02e+07          5.57e+06          2.42e+08        2.57e+08     ( 27.0%)
-----
Total         1.88e+08 pW      2.18e+07 pW      7.52e+08 pW      9.54e+08 pW
-----
1
lcc2_shell>
```

Following is qor report after placement using command: **report_qor**

Cell area = 305.23, combinational cell count = 67

Leaf cell count = 87, Buffers cell count = 7, Inverter cell count = 3

```
student@ict-chipin:~/RTL2GDSII101/ICII
File Edit View Search Terminal Tabs Help

student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x

Cell Count
-----
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 87
Buf/Inv Cell Count: 10
Buf Cell Count: 7
Inv Cell Count: 3
Combinational Cell Count: 67
  Single-bit Isolation Cell Count: 0
  Multi-bit Isolation Cell Count: 0
  Isolation Cell Banking Ratio: 0.00%
  Single-bit Level Shifter Cell Count: 0
  Multi-bit Level Shifter Cell Count: 0
  Level Shifter Cell Banking Ratio: 0.00%
  Single-bit ELS Cell Count: 0
  Multi-bit ELS Cell Count: 0
  ELS Cell Banking Ratio: 0.00%
Sequential Cell Count: 20
  Integrated Clock-Gating Cell Count: 0
  Sequential Macro Cell Count: 0
  Single-bit Sequential Cell Count: 20
  Multi-bit Sequential Cell Count: 0
  Sequential Cell Banking Ratio: 0.00%
  BitsPerFlop: 1.00
Macro Count: 0
-----

Area
-----
Combinational Area: 172.56
Noncombinational Area: 132.66
Buf/Inv Area: 23.89
Total Buffer Area: 20.08
Total Inverter Area: 3.81
Macro/Block Box Area: 0.00
Net Area: 0
Net XLength: 1025.76
Net YLength: 1413.60
-----
Cell Area (netlist): 305.23
Cell Area (netlist and physical only): 305.23
Net Length: 2439.36
-----

Design Rules
-----
Total Number of Nets: 112
Nets with Violations: 13
Max Trans Violations: 13
Max Cap Violations: 0
-----
```

Following is timing report after placement using command : **report_timing**

Slack = 0.37

```
student@ict-chipin:~/RTL2GDSII101/ICII
File Edit View Search Terminal Tabs Help

student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x

[ic2-lic Mon Jun 2 01:39:46 2025] Sending check-out request for 'ICCompilerII-NX' (1) with wait option
[ic2-lic Mon Jun 2 01:39:46 2025] Sending check-out request for 'ICCompilerII-NX' with wait option succeeded
[ic2-lic Mon Jun 2 01:39:46 2025] Sending check-out check request for 'ICCompilerII-NX'
[ic2-lic Mon Jun 2 01:39:46 2025] Check-out check request for 'ICCompilerII-NX' returned 0
[ic2-lic Mon Jun 2 01:39:46 2025] Sending count request for 'ICCompilerII-NX'
[ic2-lic Mon Jun 2 01:39:46 2025] Count request for 'ICCompilerII-NX' returned 1
[ic2-lic Mon Jun 2 01:39:46 2025] Check-out of alternate set of keys directly with queueing was successful
*****
Report : timing
  -path_type full
  -delay_type max
  -max_paths 1
  -report_by design
Design : shifter_8bit
Version : W-2024.09
Date : Mon Jun 2 01:39:46 2025
*****
Startpoint: data_out_reg[5] (rising edge-triggered flip-flop clocked by Clock)
Endpoint: data_out[5] (output port clocked by Clock)
Mode: func
Corner: nom
Scenario: func:nom
Path Group: **reg2out_default**
Path Type: max
*****
Point              Incr      Path
-----
clock Clock (rise edge)          0.00      0.00
clock network delay (propagated)  0.07      0.07
data_out_reg[5]/CLK (DFFX1_RVT)  0.00      0.07 r
data_out_reg[5]/Q (DFFX1_RVT)    0.09      0.16 f
data_out[5] (out)                 0.00      0.16 f
data arrival time                  0.16
-----
clock Clock (rise edge)          1.30      1.30
clock network delay (propagated)  0.04      1.34
clock uncertainty                 -0.30      1.04
output external delay             -0.50      0.54
data required time                 0.54
-----
data required time                 0.54
data arrival time                 -0.16
-----
slack (MET)                      0.37
*****
ic2_shell>
```

Following is the power consumption report after placement using command :

report_power

```
Report : power
- significant_digits 2
Design : shifter_2bit
Version : W-2024.09
Date : Mon Jun 2 01:40:35 2025
*****
Information: Activity for scenario func::nom was cached, no propagation required. (POW-005)
Mode: func
Corner: nom
Scenario: func::nom
Voltage: 0.85
Temperature: 125.00

Voltage Unit : 1V
Capacitance Unit : 1fF
Time Unit : 1ns
Temperature Unit : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW

Switched supply net power scaling:
scaling for Leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

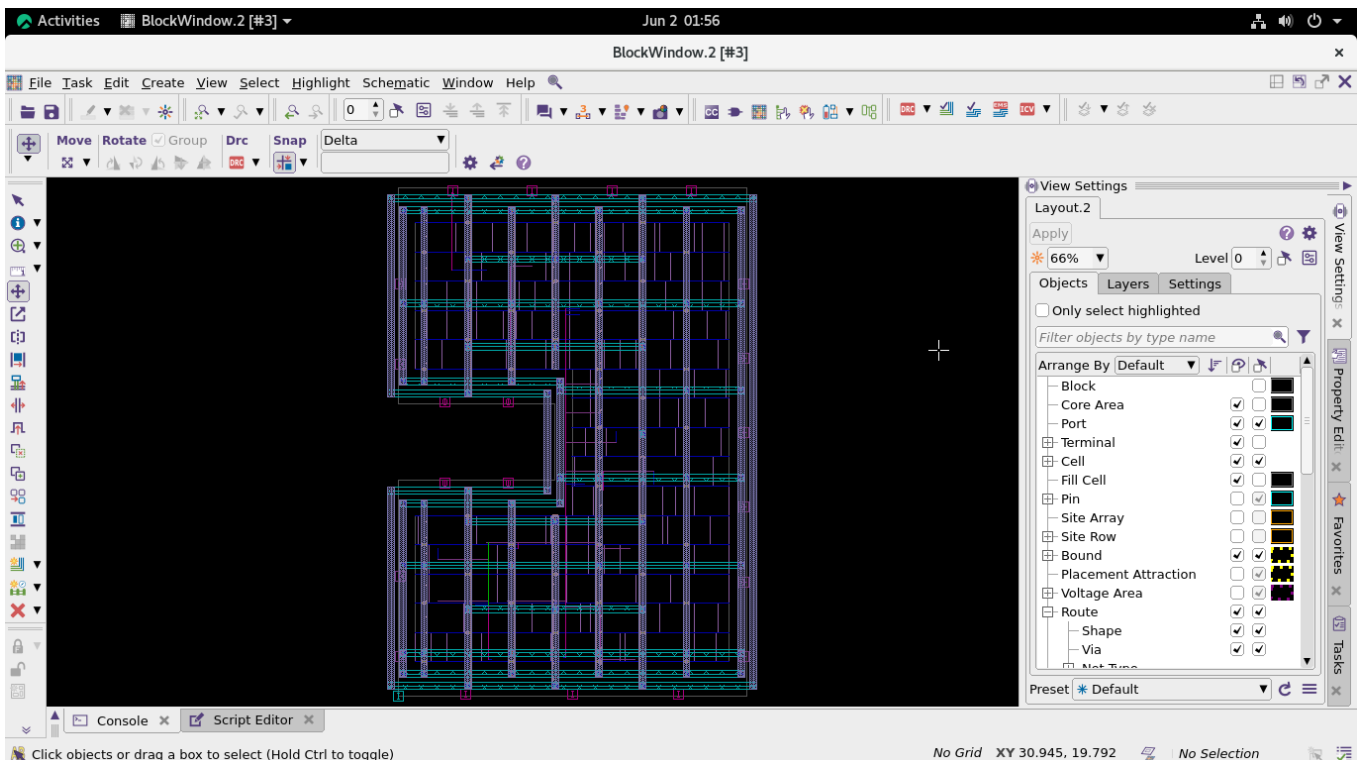
Cell Internal Power = 1.88e+08 pW ( 89.2%)
Net Switching Power = 2.18e+07 pW ( 10.8%)
Total Dynamic Power = 2.02e+08 pW (100.0%)
Cell Leakage Power = 7.52e+08 pW

Attributes
-----
u - User defined power group
i - Includes clock pin internal power

Power Group Internal Power Switching Power Leakage Power Total Power ( % ) Attrs
-----
io_pad 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
memory 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
black_box 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
clock_network 2.04e+08 1.31e+07 9.75e+07 3.15e+08 ( 33.0%) i
register -3.40e+07 3.10e+06 4.13e+08 3.82e+08 ( 40.1%)
sequential 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
combinational 1.02e+07 5.57e+06 2.42e+08 2.57e+08 ( 27.0%)
-----
Total 1.88e+08 pW 2.18e+07 pW 7.52e+08 pW 9.54e+08 pW
i
icc2_shell>
```

Now we did is what we call CTS (Clock Time Synthesis) which is done to distribute the clock signal to all sequential elements (like flip-flops) efficiently and uniformly across the chip after placement but before routing.

Run the command to source clock.tcl file: **source scripts/clock.tcl** , we will get this



We can see some magenta and green connection lines, green lines means routed segments of clock tree and magenta lines are for virtual connections that still needed to be routed.

Following is qor report after CTS using command: **report_qor**

Cell area = 309.29, combinational cell count = 69

Leaf cell count = 89, Buffers cell count = 9, Inverter cell count = 3

```
Activities Terminal Jun 2 01:59
student@ict-chipin:~/RTL2GDSII101/ICII

File Edit View Search Terminal Tabs Help

student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi...

Cell Count
-----
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 89
Buf/Inv Cell Count: 12
Buf Cell Count: 9
Inv Cell Count: 3
Combinational Cell Count: 69
  Single-bit Isolation Cell Count: 0
  Multi-bit Isolation Cell Count: 0
  Isolation Cell Banking Ratio: 0.00%
  Single-bit Level Shifter Cell Count: 0
  Multi-bit Level Shifter Cell Count: 0
  Level Shifter Cell Banking Ratio: 0.00%
  Single-bit ELS Cell Count: 0
  Multi-bit ELS Cell Count: 0
  ELS Cell Banking Ratio: 0.00%
Sequential Cell Count: 20
  Integrated Clock-Gating Cell Count: 0
  Sequential Macro Cell Count: 0
  Single-bit Sequential Cell Count: 20
  Multi-bit Sequential Cell Count: 0
  Sequential Cell Banking Ratio: 0.00%
  BitsPerFlop: 1.00
Macro Count: 0

Area
-----
Combinational Area: 176.63
Noncombinational Area: 132.66
Buf/Inv Area: 27.96
Total Buffer Area: 24.14
Total Inverter Area: 3.81
Macro/Block Box Area: 0.00
Net Area: 0
Net Mlength: 1033.10
Net Vlength: 1387.63

Cell Area (netlist): 309.29
Cell Area (netlist and physical only): 309.29
Net Length: 2420.94

Design Rules
-----
Total Number of Nets: 114
Nets with Violations: 13
Max Trans Violations: 13
Max Cap Violations: 0
```

Following is timing report after CTS using command : **report_timing**

Slack = 0.37

```
Activities Terminal Jun 2 02:00
student@ict-chipin:~/RTL2GDSII101/ICII

File Edit View Search Terminal Tabs Help

student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi...

[iccd2-lic Mon Jun 2 02:00:16 2025] Sending check-out request for "ICCompiler-II-NX" (1) with wait option
[iccd2-lic Mon Jun 2 02:00:16 2025] Checking status for feature ICCompilerII-NX
[iccd2-lic Mon Jun 2 02:00:16 2025] Client:ict-chipin.sot.pdu.ac.in Server:27020@c2s.cdacb.in Checkout succeeded ICCompilerII-NX 2024.09
[iccd2-lic Mon Jun 2 02:00:16 2025] Check-out request for "ICCompilerII-NX" with wait option succeeded
[iccd2-lic Mon Jun 2 02:00:16 2025] Sending checkout check request for "ICCompilerII-NX"
[iccd2-lic Mon Jun 2 02:00:16 2025] Checkout check request for "ICCompilerII-NX" returned 0
[iccd2-lic Mon Jun 2 02:00:16 2025] Sending count request for "ICCompilerII-NX"
[iccd2-lic Mon Jun 2 02:00:16 2025] Count request for "ICCompilerII-NX" returned 1
[iccd2-lic Mon Jun 2 02:00:16 2025] Check-out of alternate set of keys directly with queueing was successful

Report : Timing
  -path_type full
  -delay_type max
  -max_paths 1
  -report_by design
Design : shifter_8bit
Version: M 2024.09
Date : Mon Jun 2 02:00:16 2025
=====
Startpoint: data_out_reg[5] (rising edge-triggered flip-flop clocked by Clock)
Endpoint: data_out[5] (output port clocked by Clock)
Mode: func
Corner: nom
Scenario: func:nom
Path Group: **reg2out_default**
Path Type: max

Point                                Incr      Path
-----
clock Clock (rise edge)              0.00      0.00
clock network delay (propagated)      0.07      0.07
data_out_reg[5]/CLK (DFFX1_RVT)      0.00      0.07 f
data_out_reg[5]/Q (DFFX1_RVT)        0.00      0.17 f
data_out[5] (out)                    0.00      0.17
data arrival time                     0.17
clock Clock (rise edge)              1.30      1.30
clock network delay (propagated)      0.04      1.34
clock uncertainty                     -0.30      1.04
output external delay                 -0.50      0.54
data required time                    0.54
data arrival time                    -0.17
slack (MET)                          0.37

iccd2_shell>
```

Following is the power consumption report after CTS using command :

report_power

```
Activities Terminal Jun 2 02:03
student@ict-chipin:~/RTL2GDSII101/IC11

File Edit View Search Terminal Tabs Help

student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi... student@ict-chi...

Information: Activity propagation will be performed for scenario func::nom.
Information: Doing activity propagation for mode 'func' and corner 'nom' with effort level 'medium'. (P0W-024)
Information: Timer-derived activity data is cached on scenario func::nom (P0W-022)
Information: Fast mode activity propagation power.rtl_activity_annotation setup is ignored. Always use accurate mode.
Information: Running switching activity propagation in scalar mode!

**** Information : No. of simulation cycles = 6 ****
Mode: func
Corner: nom
Scenario: func::nom
Voltage: 0.85
Temperature: 125.00

Voltage Unit : 1V
Capacitance Unit : 1fF
Time Unit : 1ns
Temperature Unit : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW

Switched supply net power scaling:
scaling for Leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

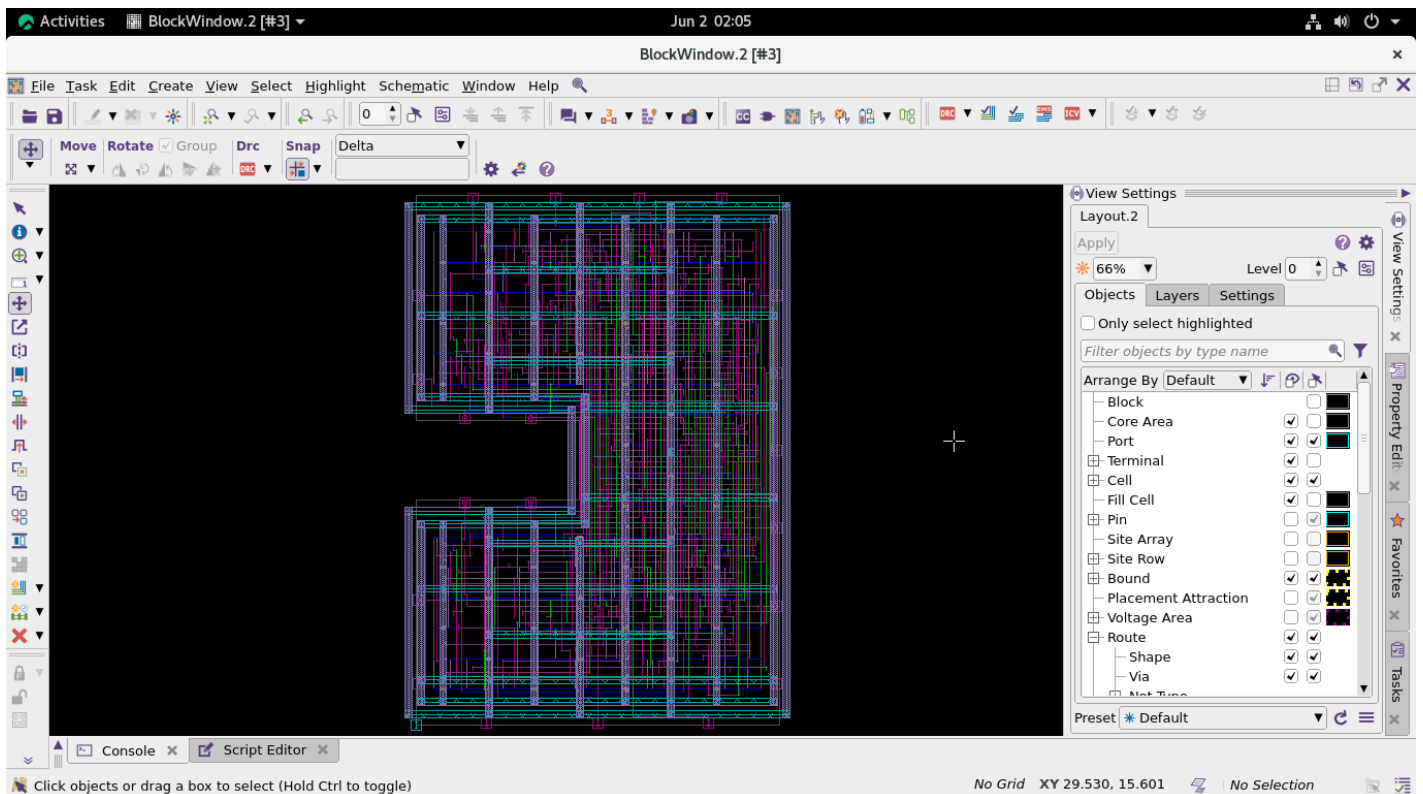
Cell Internal Power = 1.82e+08 pW ( 88.3%)
Net Switching Power = 2.42e+07 pW ( 11.7%)
Total Dynamic Power = 2.06e+08 pW (100.0%)
Cell Leakage Power = 7.68e+08 pW

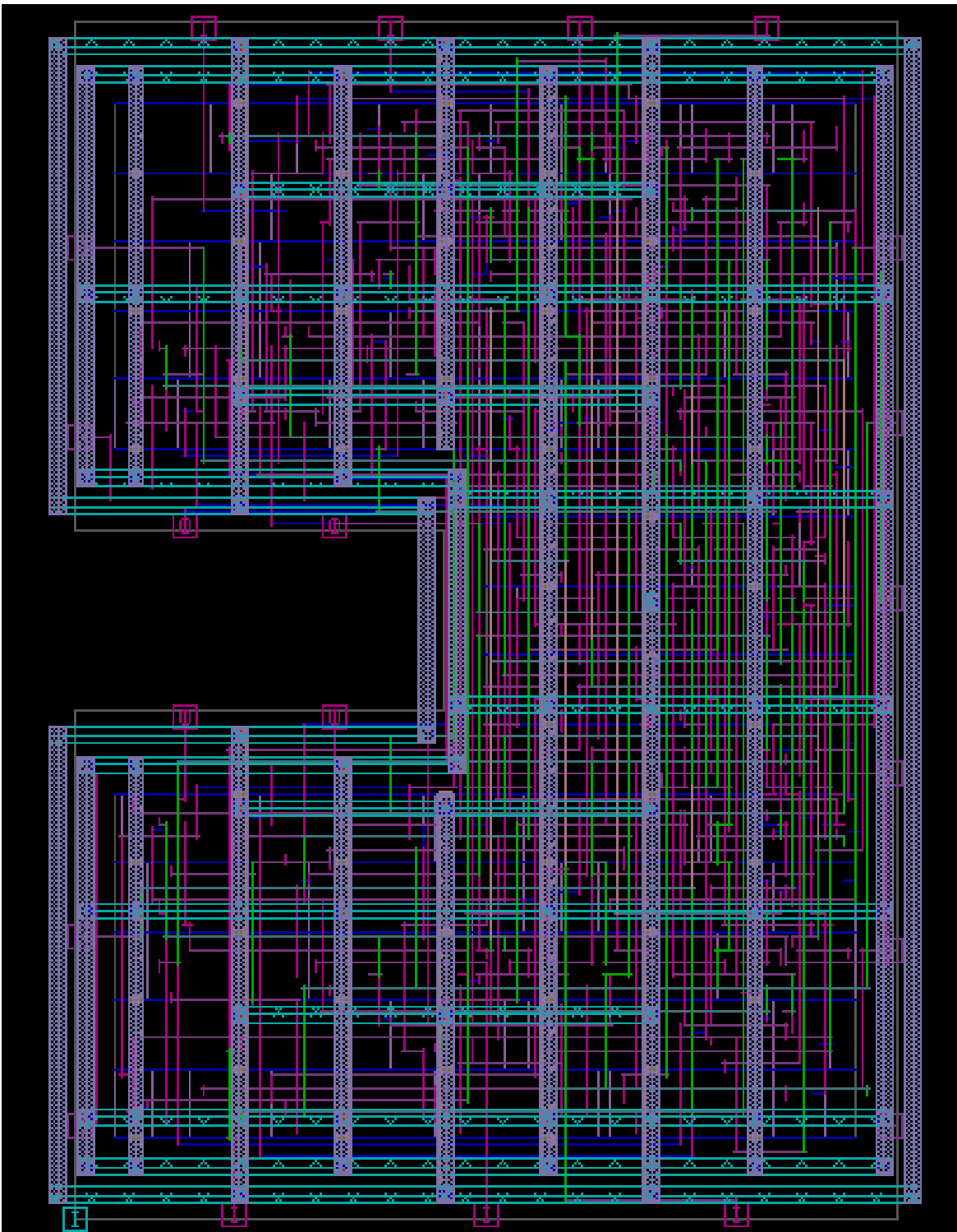
Attributes
-----
u - User defined power group
i - Includes clock pin internal power

Power Group Internal Power Switching Power Leakage Power Total Power ( % ) Attrs
-----
io_pad 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
memory 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
black_box 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
clock_network 2.04e+08 1.38e+07 9.75e+07 3.15e+08 ( 32.4%) i
register -3.23e+07 3.54e+06 4.13e+08 3.84e+08 ( 39.5%)
sequential 0.00e+00 0.00e+00 0.00e+00 0.00e+00 ( 0.0%)
combinational 1.02e+07 6.83e+06 2.58e+08 2.75e+08 ( 28.2%)
-----
Total 1.82e+08 pW 2.42e+07 pW 7.68e+08 pW 9.75e+08 pW
i
scc2_shell>
```

Now the last stage is routing which connects all logic elements (cells, blocks, IOs) with metal wires based on the netlist. We need this to create physical paths for signals and clocks so the chip functions correctly.

Run the command to source route.tcl file: **source scripts/route.tcl** , we will get this





Above figure is magnified routed layout showing a routed digital design where metal wires (green/magenta) connect the placed standard cells (blue blocks). It represents the interconnect fabric of your circuit, post-CTS and after routing.

Following is qor report after routing using command: **report_qor**

Cell area = 309.04, combinational cell count = 69

Leaf cell count = 89, Buffers cell count = 9, Inverter cell count = 3

```
Activities Terminal Jun 2 02:08
student@ict-chipin:~/RTL2GDSII101/ICII

File Edit View Search Terminal Tabs Help

student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x

Cell Count
-----
Hierarchical Cell Count: 0
Hierarchical Port Count: 0
Leaf Cell Count: 89
Buf/Inv Cell Count: 12
Buf Cell Count: 9
Inv Cell Count: 3
Combinational Cell Count: 69
  Single-bit Isolation Cell Count: 0
  Multi-bit Isolation Cell Count: 0
  Isolation Cell Banking Ratio: 0.00%
  Single-bit Level Shifter Cell Count: 0
  Multi-bit Level Shifter Cell Count: 0
  Level Shifter Cell Banking Ratio: 0.00%
  Single-bit ELS Cell Count: 0
  Multi-bit ELS Cell Count: 0
  ELS Cell Banking Ratio: 0.00%
Sequential Cell Count: 20
  Integrated Clock-Gating Cell Count: 0
  Sequential Macro Cell Count: 0
  Single-bit Sequential Cell Count: 20
  Multi-bit Sequential Cell Count: 0
  Sequential Cell Banking Ratio: 0.00%
  BitsPerFlop: 1.00
Macro Count: 0
-----

Area
-----
Combinational Area: 176.38
Noncombinational Area: 132.66
Buf/Inv Area: 27.96
Total Buffer Area: 24.14
Total Inverter Area: 3.81
Macro/Black Box Area: 0.00
Net Area: 0
Net XLength: 1125.85
Net YLength: 1475.14
-----
Cell Area (netlist): 309.04
Cell Area (netlist and physical only): 309.04
Net Length: 2691.00
-----

Design Rules
-----
Total Number of Nets: 114
Nets with Violations: 13
Max Trans Violations: 13
Max Cap Violations: 0
```

Following is timing report after routing using command : **report_timing**

Slack = 0.37

```
Activities Terminal Jun 2 02:09
student@ict-chipin:~/RTL2GDSII101/ICII

File Edit View Search Terminal Tabs Help

student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x student@ict-chi... x

[ic2-lic Mon Jun 2 02:09:50 2025] Sending check-out request for 'ICCompilerII-NX' (1) with wait option
[SC1] 06/02/2025 02:09:50 Checking status for feature ICCompilerII-NX
[SC1] 06/02/2025 02:09:50 P10:393 Client:ict-chipin.set.pdu.ac.in Server:27020@c2s.cdac.in Checkout succeeded ICCompilerII-NX 2024.09
[ic2-lic Mon Jun 2 02:09:50 2025] Check-out request for 'ICCompilerII-NX' with wait option succeeded
[ic2-lic Mon Jun 2 02:09:50 2025] Sending checkout check request for 'ICCompilerII-NX'
[ic2-lic Mon Jun 2 02:09:50 2025] Checkout check request for 'ICCompilerII-NX' returned 0
[ic2-lic Mon Jun 2 02:09:50 2025] Sending count request for 'ICCompilerII-NX'
[ic2-lic Mon Jun 2 02:09:50 2025] Count request for 'ICCompilerII-NX' returned 1
[ic2-lic Mon Jun 2 02:09:50 2025] Check-out of alternate set of keys directly with queueing was successful
*****
Report : timing
  -path_type full
  -delay_type max
  -max_paths 1
  -report_by design
Design : shifter_8bit
Version: W-2024.09
Date : Mon Jun 2 02:09:50 2025
*****
Startpoint: data_out_reg[2] (rising edge-triggered flip-flop clocked by Clock)
Endpoint: data_out[2] (output port clocked by Clock)
Mode: func
Corner: nom
Scenario: func:nom
Path Group: **reg2out_default**
Path Type: max

Point                               Incr      Path
-----
clock Clock (rise edge)              0.00      0.00
clock network delay (propagated)      0.07      0.07

data_out_reg[2]/CLK (DFFX1_RVT)      0.00      0.07 r
data_out_reg[2]/Q (DFFX1_RVT)        0.09      0.17 f
data_out[2] (out)                     0.00      0.17 f
data arrival time                     0.17

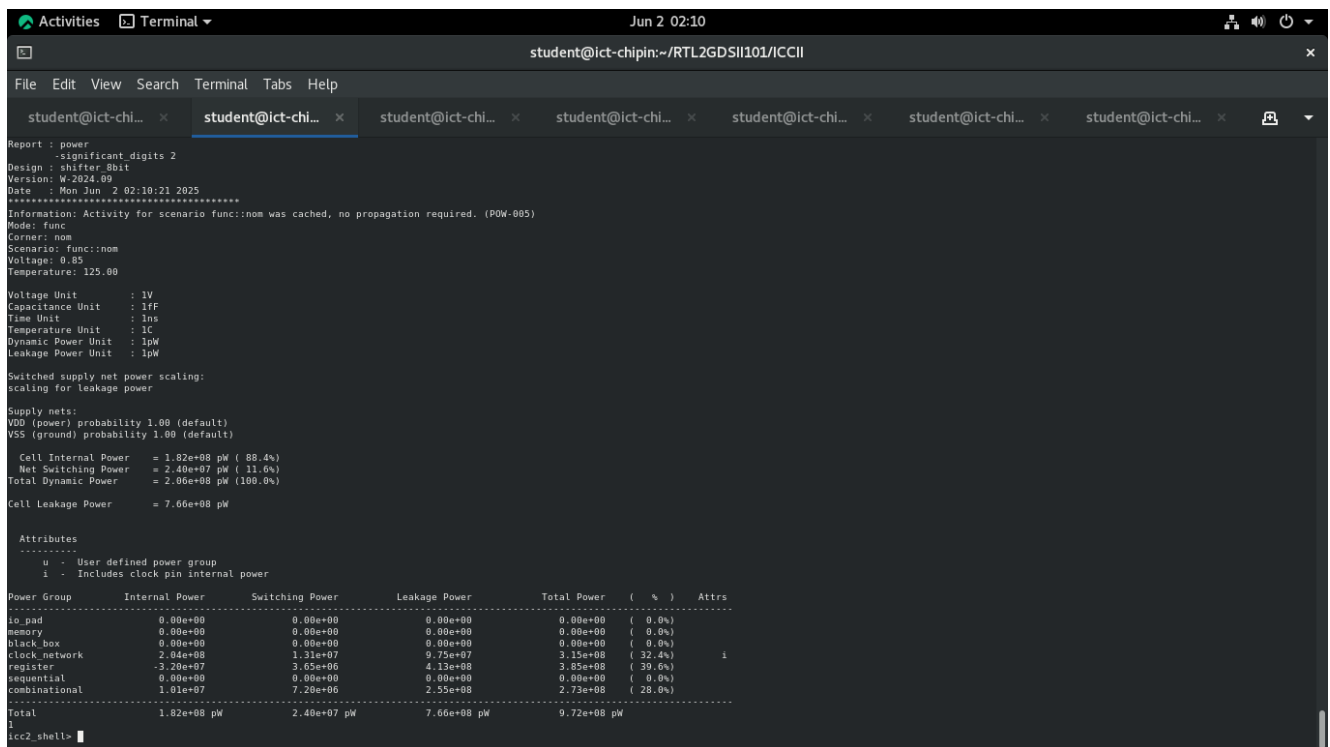
clock Clock (rise edge)              1.30      1.30
clock network delay (propagated)      0.04      1.34
clock uncertainty                     -0.30      1.04
output external delay                 -0.50      0.54
data required time                    0.54

data required time                    0.54
data arrival time                     -0.17
-----
slack (MET)                           0.37

1
ic2_shell>
```

Following is the power consumption report after routing using command :

report_power



```
Report : power
- significant_digits 2
Design : shifter 8bit
Version: W-2024.09
Date   : Mon Jun 2 02:10:21 2025
*****
Information: Activity for scenario func::nom was cached, no propagation required. (POW-005)
Mode: func
Corner: nom
Scenario: func::nom
Voltage: 0.85
Temperature: 125.00

Voltage Unit      : 1V
Capacitance Unit : 1fF
Time Unit        : 1ns
Temperature Unit  : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)

Cell Internal Power   = 1.82e+08 pW ( 88.4%)
Net Switching Power   = 2.40e+07 pW ( 11.6%)
Total Dynamic Power   = 2.06e+08 pW (100.0%)
Cell Leakage Power    = 7.66e+08 pW

Attributes
-----
u - User defined power group
i - Includes clock pin internal power

Power Group      Internal Power      Switching Power      Leakage Power      Total Power      ( % )      Attrs
-----
io_pad           0.00e+00      0.00e+00      0.00e+00      0.00e+00      ( 0.0%)
memory          0.00e+00      0.00e+00      0.00e+00      0.00e+00      ( 0.0%)
black_box       0.00e+00      0.00e+00      0.00e+00      0.00e+00      ( 0.0%)
clock_network   2.04e+08      1.31e+07      9.75e+07      3.15e+08      ( 32.4%)      i
register        3.20e+07      3.65e+06      4.13e+08      3.55e+08      ( 30.4%)
sequential      0.00e+00      0.00e+00      0.00e+00      0.00e+00      ( 0.0%)
combinational   1.01e+07      7.20e+06      2.55e+08      2.73e+08      ( 28.0%)
-----
Total           1.82e+08 pW      2.40e+07 pW      7.66e+08 pW      9.72e+08 pW

i
icc2_shell>
```

Now PrimeTime for Static Timing Analysis (STA) which is a critical step in digital design to verify timing after synthesis, placement, and routing.

PrimeTime is used to accurately check timing of a chip without simulation, ensuring the design will work reliably at the target clock frequency.

In .sdc file in CONSTRAINTS folder I gave 1.3ns clock period so the clock frequency will be:

$$\begin{aligned}\text{Clock Frequency} &= 1 / \text{clock period} \\ &= 1 / 1.3 \times 10^{-9} \\ &= 769.23 \text{ MHz}\end{aligned}$$

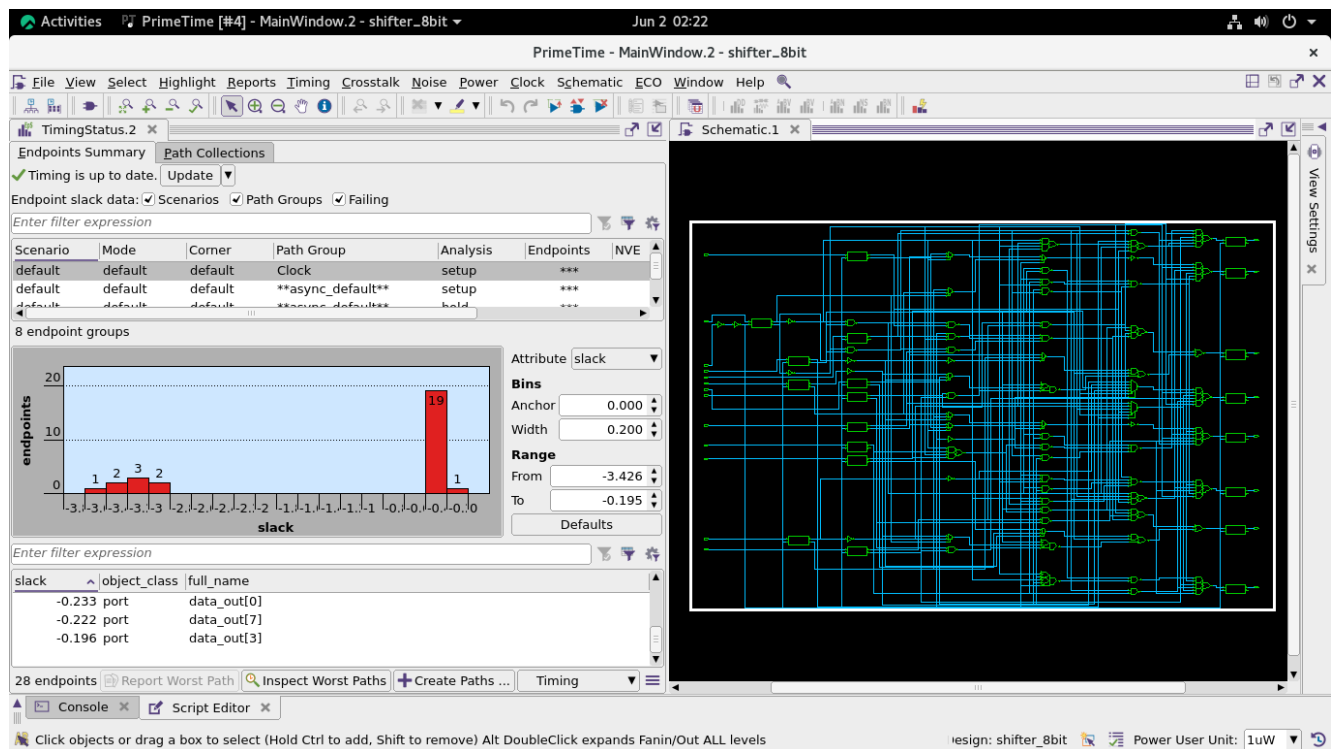
First go into PT folder through terminal and there we can see 2 folders:

1. run_pt_p1.tcl
2. run_pt_p2.tcl

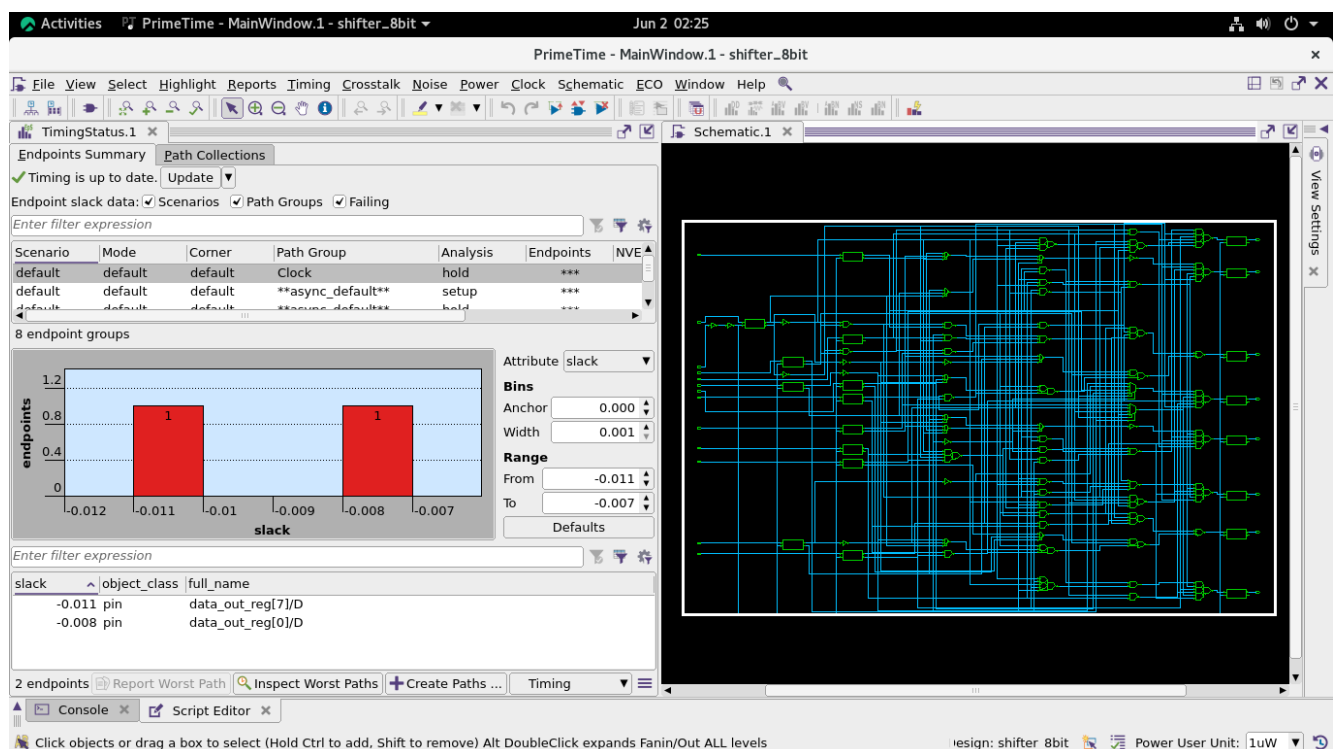
In PT folder open the primetime shell using command: **pt_shell**

Now source any file from 2 files and open GUI of primetime.

Command in terminal: `source run_pt_p1.tcl`

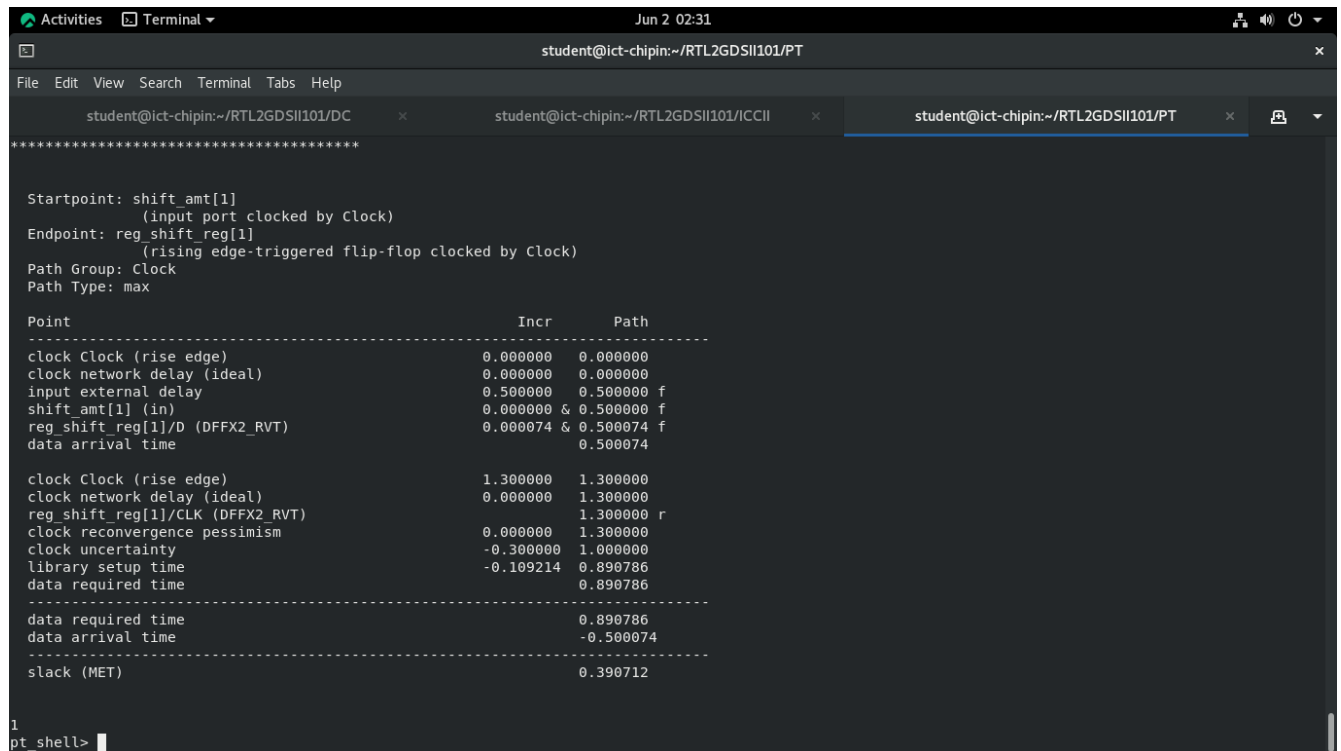


Command in terminal: `source run_pt_p2.tcl`



PrimeTime GUI shows a histogram of the worst-case slacks at all endpoints.

When we run command in pt_shell → **report_timing** , considering run_pt_p2.tcl file i get slack = 0.39



```
*****
Startpoint: shift_amt[1]
              (input port clocked by Clock)
Endpoint: reg_shift_reg[1]
              (rising edge-triggered flip-flop clocked by Clock)
Path Group: Clock
Path Type: max

Point                                     Incr      Path
-----
clock Clock (rise edge)                   0.000000   0.000000
clock network delay (ideal)               0.000000   0.000000
input external delay                      0.500000   0.500000 f
shift_amt[1] (in)                         0.000000 & 0.500000 f
reg_shift_reg[1]/D (DFFX2_RVT)           0.000074 & 0.500074 f
data arrival time                         0.500074

clock Clock (rise edge)                   1.300000   1.300000
clock network delay (ideal)               0.000000   1.300000
reg_shift_reg[1]/CLK (DFFX2_RVT)          1.300000 r
clock reconvergence pessimism             0.000000   1.300000
clock uncertainty                         -0.300000   1.000000
library setup time                       -0.109214   0.890786
data required time                        0.890786

-----
data required time                        0.890786
data arrival time                        -0.500074
-----
slack (MET)                              0.390712

1
pt_shell>
```

They are different because the PrimeTime GUI shows the worst-case slack across all endpoints, while report_timing in pt_shell shows slack for a specific timing path.

Learnings and Conclusion :

1. Understanding of Verilog HDL and RTL Design

Through the implementation of an 8-bit shifter that shifts left or right by N bits, I gained a strong understanding of writing Register Transfer Level (RTL) code in Verilog. I also learned how to structure and test digital circuits using Verilog modules and testbenches.

2. Simulation and Debugging Skills

By simulating the design using vcs and visualizing waveforms in Verdi, I improved my debugging skills. Observing functional correctness for different test cases helped reinforce the importance of simulation before synthesis.

3. Knowledge of Synthesis Using Synopsys Design Compiler (DC)

I learned how to use Synopsys DC for logic synthesis by modifying TCL scripts and constraints. I understood how to analyze the synthesized netlist and how timing constraints affect design performance. The concept of timing slack and its impact on the design's efficiency became clear through practical implementation.

4. Hands-on Experience with ICC2 for Physical Design

From floorplanning to routing, I explored each stage of digital backend design using Synopsys ICC2. I experimented with different floorplan scenarios and understood how layout quality impacts area, power, and performance. Concepts like placement, power planning, CTS, and routing were clearly understood through this practical approach.

5. Static Timing Analysis with PrimeTime

I performed static timing analysis (STA) using PrimeTime and understood the significance of slack, endpoint timing paths, and worst-case analysis. This step taught me how to validate that a chip design meets its timing requirements post-layout.

6. Power and Area Estimation

I learned how to analyze power consumption and cell area at various design stages. I observed how synthesis and physical layout steps affect these metrics and how optimization is necessary to meet design goals.

7. Integration of Complete ASIC Design Flow

Overall, I gained a comprehensive understanding of the complete ASIC design flow from RTL coding and simulation to physical design and timing closure. This hands-on experience strengthened my foundation in VLSI design and provided practical exposure to industry tools and flows.